O*NET OnLine

Software Developers

15-1252.00

Bright Outlook | Updated 2026

Research, design, and develop computer and network software or specialized utility programs. Analyze user needs and develop software solutions, applying principles and techniques of computer science, engineering, and mathematical analysis. Update software or enhance existing software capabilities. May work with computer hardware engineers to integrate hardware and software systems, and develop specifications and performance requirements. May maintain databases within an application area, working individually or coordinating database development as part of a team.

Sample of reported job titles: Application Developer, Application Integration Engineer, Developer, DevOps Engineer (Development Operations Engineer), Infrastructure Engineer, Software Architect, Software Developer, Software Development Engineer, Software Engineer, Systems Engineer

Occupation-Specific Information

Tasks

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- Analyze user needs and software requirements to determine feasibility of design within time and cost constraints.
- Develop or direct software system testing or validation procedures, programming, or documentation.
- Confer with systems analysts, engineers, programmers and others to design systems and to obtain information on project limitations and capabilities, performance requirements and interfaces.
- Modify existing software to correct errors, adapt it to new hardware, or upgrade interfaces and improve performance.
- Prepare reports or correspondence concerning project specifications, activities, or status.

Occupational Requirements

Work Activities

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- **Working with Computers** — Using computers and computer systems (including hardware and software) to program, write software, set up functions, enter data, or process information.
- **Processing Information** — Compiling, coding, categorizing, calculating, tabulating, auditing, or verifying information or data.
- **Making Decisions and Solving Problems** — Analyzing information and evaluating results to choose the best solution and solve problems.
- **Thinking Creatively** — Developing, designing, or creating new applications, ideas, relationships, systems, or products, including artistic contributions.
- **Communicating with Supervisors, Peers, or Subordinates** — Providing information to supervisors, co-workers, and subordinates by telephone, in written form, e-mail, or in person.

Detailed Work Activities

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- Monitor computer system performance to ensure proper operation.
- Analyze project data to determine specifications or requirements.
- Develop testing routines or procedures.
- Manage information technology projects or system activities.
- Modify software programs to improve performance.

Work Context

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- **Spend Time Sitting** — 95% responded “Continually or almost continually.”
- **Work With or Contribute to a Work Group or Team** — 71% responded “Extremely important.”
- **E-Mail** — 13% responded “Once a year or more but not every month.”
- **Freedom to Make Decisions** — 46% responded “Some freedom.”
- **Importance of Being Exact or Accurate** — 44% responded “Very important.”

Experience Requirements

Job Zone

Title	Job Zone Four: Considerable Preparation Needed
Education	Most of these occupations require a four-year bachelor's degree, but some do not.
Related Experience	A considerable amount of work-related skill, knowledge, or experience is needed for these occupations. For example, an accountant must complete four years of college and work for several years in accounting to be considered qualified.
Job Training	Employees in these occupations usually need several years of work-related experience, on-the-job training, and/or vocational training.

Job Zone Examples	Many of these occupations involve coordinating, supervising, managing, or training others. Examples include real estate brokers, sales managers, database administrators, graphic designers, conservation scientists, art directors, and cost estimators.
SVP Range	(7.0 to < 8.0)

Training & Credentials

State training Select a State Go

Local training ZIP Code Go

Certifications Find certifications nationwide

State licenses Select a State Go

Apprenticeship Opportunities

All 4 displayed

Example apprenticeship titles for this occupation:

- Application Developer
- Commercial Drone Software Developer
- Devops Engineer (Nof)
- Software Developer (Nof)

Specific title(s) listed above are vetted by industry and approved by the U.S. Department of Labor for use in a Registered Apprenticeship Program.

Start your career and build your skillset. Visit **Apprenticeship.gov** to learn about opportunities related to this occupation.

Worker Requirements

Software Skills

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- **Data base user interface and query software** — Airtable; Blackboard software; GraphQL; Prometheus
- **Development environment software** — Apache Kafka; Apache Maven; Oracle Java 2 Platform Enterprise Edition J2EE; Oracle SQL Developer
- **Object or component oriented development software** — Apache Spark; jQuery; Scala; TypeScript

- **Operating system software** — Apple iOS ; Google Android ; Microsoft Windows Server ; UNIX Shell
- **Web platform development software** — Bootstrap ; React ; Spring Framework ; Vue.js

Hot Technologies are requirements most frequently included across all employer job postings.
See all 152 Hot Technologies for this occupation.

In Demand skills are frequently included in employer job postings for this occupation.
See all 30 In Demand skills for this occupation.

Essential Skills

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- **Critical Thinking** — Using logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions, or approaches to problems.
- **Active Learning** — Understanding the implications of new information for both current and future problem-solving and decision-making.
- **Reading Comprehension** — Understanding written sentences and paragraphs in work-related documents.
- **Active Listening** — Giving full attention to what other people are saying, taking time to understand the points being made, asking questions as appropriate, and not interrupting at inappropriate times.
- **Writing** — Communicating effectively in writing as appropriate for the needs of the audience.

Transferable Skills

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- **Programming** — Writing computer programs for various purposes.
- **Judgment and Decision Making** — Considering the relative costs and benefits of potential actions to choose the most appropriate one.
- **Systems Analysis** — Determining how a system should work and how changes in conditions, operations, and the environment will affect outcomes.
- **Complex Problem Solving** — Identifying complex problems and reviewing related information to develop and evaluate options and implement solutions.
- **Systems Evaluation** — Identifying measures or indicators of system performance and the actions needed to improve or correct performance, relative to the goals of the system.

Knowledge

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- **Computers and Electronics** — Knowledge of circuit boards, processors, chips, electronic equipment, and computer hardware and software, including applications and programming.
- **Customer and Personal Service** — Knowledge of principles and processes for providing customer and personal services. This includes customer needs assessment, meeting quality standards for services, and evaluation of customer satisfaction.

- **Mathematics** — Knowledge of arithmetic, algebra, geometry, calculus, statistics, and their applications.
- **English Language** — Knowledge of the structure and content of the English language including the meaning and spelling of words, and rules of composition and grammar.

Education

How much education does a new hire need to perform a job in this occupation? Respondents said:

85%	Bachelor's degree required
5%	Associate's degree or other 2-year degree required
5%	Master's degree required

Worker Characteristics

Abilities

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- **Deductive Reasoning** — The ability to apply general rules to specific problems to produce answers that make sense.
- **Near Vision** — The ability to see details at close range (within a few feet of the observer).
- **Oral Comprehension** — The ability to listen to and understand information and ideas presented through spoken words and sentences.
- **Oral Expression** — The ability to communicate information and ideas in speaking so others will understand.
- **Problem Sensitivity** — The ability to tell when something is wrong or is likely to go wrong. It does not involve solving the problem, only recognizing that there is a problem.

Interests

All 2 displayed

Interest code: **IC**

Want to discover your interests? Take the **O*NET Interest Profiler**.

- **Investigative** — Work involves studying and researching non-living objects, living organisms, disease or other forms of impairment, or human behavior. Investigative occupations are often associated with physical, life, medical, or social sciences, and can be found in the fields of humanities, mathematics/statistics, information technology, or health care service.
  - **Mathematics/Statistics** — Work involves developing and applying mathematics and statistical theory and methods.
- **Conventional** — Work involves following procedures and regulations to organize information or data, typically in a business setting. Conventional occupations are often associated with office work, accounting, mathematics/statistics, information technology, finance, or human resources.

- **Information Technology** — Work involves designing, developing, testing, and maintaining computer software, hardware, networks, and systems.
- **Mathematics/Statistics** — Work involves developing and applying mathematics and statistical theory and methods.

Work Styles

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- **Innovation** — A tendency to be inventive, to be imaginative, and to adopt new perspectives on ways to accomplish work.
- **Adaptability** — A tendency to be open to and comfortable with change, new experiences, or ideas at work.
- **Achievement Orientation** — A tendency to establish and maintain personally challenging work-related goals, set high work-related standards, and exert high effort toward meeting those goals and standards.
- **Intellectual Curiosity** — A tendency to seek out and acquire new work-related knowledge and obtain a deep understanding of work-related subjects.
- **Attention to Detail** — A tendency to be detail-oriented, organized, and thorough in completing work.

Workforce Characteristics

Wages & Employment Trends

Median wages (2025)	$65.38 hourly, $135,980 annual
State wages	Select a State Go
Local wages	ZIP Code Go
Employment (2025)	1,717,800 employees
Projected growth (2025-2035)	Much faster than average (7% or higher)
Projected job openings (2025-2035)	95,300
State trends	Select a State Go
Top industries (2025)	Professional, Scientific, and Technical Services Information

Source: Bureau of Labor Statistics 2025 wage data and 2025-2035 employment projections. "Projected growth" represents the estimated change in total employment over the projections period (2025-2035). "Projected job openings" represent openings due to growth and replacement.

Job Openings on the Web

State job openings Select a State Go

Local job openings ZIP Code Go

More Information

Related Occupations

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- 15-1251.00 Computer Programmers
- 15-1211.00 Computer Systems Analysts
- 15-1299.08 Computer Systems Engineers/Architects
- 15-1242.00 Database Administrators
- 15-1253.00 Software Quality Assurance Analysts and Testers

Professional Associations

Disclaimer: Sources are listed to provide additional information on related jobs, specialties, and/or industries. Links to non-DOL Internet sites are provided for your convenience and do not constitute an endorsement.

National Associations

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- Association for Computing Machinery
- Association for Information Science and Technology
- Association for Information Systems
- Association for the Advancement of Artificial Intelligence
- Association for Women in Computing

Regional Associations

- Midwest Association for Information Systems
- Southern Association for Information Systems

Accreditation, Certification, & Unions

- CompTIA
- Project Management Institute